Approximating Total Variation Distance Between Product Distributions

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I. Abstract

Total variation (TV) distance between product distributions is $\#\mathbf{P}$-hard to compute exactly, motivating the study of efficient approximation algorithms. This paper surveys three lines of work on this problem and contributes both an adaptive sampling improvement to the randomized algorithm and an empirical analysis of the deterministic FPTAS.

We first present the randomized FPRAS of Feng et al., which achieves runtime $O(n^2\varepsilon^{-2}\log\delta^{-1})$ via an importance-sampling scheme over a greedy coupling. We then introduce an *adaptive Monte Carlo stopping rule* that replaces the fixed worst-case batch size $m = \lceil 10n/\varepsilon^2 \rceil$ with a data-driven criterion based on the empirical coefficient of variation, tracked online via Welford's algorithm. Experiments on Bernoulli product distributions confirm that the (ε, δ)-approximation guarantees are preserved across 270 independent runs with zero failures, while per-batch sample counts become effectively independent of n for well-separated distributions. Speedups over the fixed-sample baseline scale linearly with n, reaching $17.3\times$ at $n = 13$ under $\varepsilon = 0.5$, and up to $169.2\times$ in the $\varepsilon = 0.1$ regime where the adaptive criterion has the most room to operate.

We then show that the FPRAS framework extends fully to continuous product distributions over $\mathbb{R}^n$ under natural oracle assumptions, resolving a genuine gap in the original framework through the non-jointly-continuous structure of the maximal coupling. We then cover the deterministic FPTAS of Feng, Liu, and Liu, which achieves runtime $O(qn^2/\varepsilon \cdot \log q \cdot \log(n/\varepsilon \cdot \mathrm{TV}))$ by treating the likelihood ratio as a probability distribution and introducing the minimum total variation distance as a metric for sparsification error.

Experiments on the deterministic algorithm across five distribution families yield three findings beyond what theory predicts.

II. Introduction

The total variation distance between P and Q over finite Ω is $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_x |P(x) - Q(x)| = \sum_x \max(0, P(x) - Q(x))$. It equals the maximum probability bias of any event between the two distributions and is the canonical metric in probability and statistics.

For product distributions $P = \bigotimes_i P_i$ and $Q = \bigotimes_i Q_i$ over $[q]^n$, a direct computation requires summing over all q^n outcomes. Bhattacharyya et al. [3] proved this is $\#\mathbf{P}$-complete even over the Boolean domain. This separation from distances such as KL, Hellinger, and χ^2 — which tensorise over marginals and are computable in $O(nq)$ — motivates the approximation algorithms studied here.

This paper has three parts. III presents the randomized FPRAS of [2] and its key lemmas. V presents a team extension of that algorithm to continuous product distributions, which resolves a genuine gap in the original framework. VI covers the deterministic restricted algorithm of [3] and the general deterministic FPTAS of [1] in depth. VII gives empirical analysis of the deterministic algorithm across five distribution families.

III. Randomized Algorithm for Product Distributions

A. Algorithm Setup

The algorithm of [2] works with product distributions over $[q]^n$. Rather than using the optimal coupling $\mathcal{O}$ of P and Q (which requires global knowledge), it uses the *coordinate-wise greedy coupling* $\mathcal{C} = \mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_n$, where each $\mathcal{C}_i$ couples P_i and Q_i optimally and independently. Since coordinates are independent under $\mathcal{C}$:

$$\mathbb{P}_{\mathcal{C}}[X \neq Y] = 1 - \prod_{i=1}^{n}(1 - \mathrm{TV}(P_i, Q_i)),$$

which is computable in $O(n)$ time. Any coupling satisfies $\mathrm{TV}(P,Q) \leq \mathbb{P}_{\mathcal{C}}[X \neq Y]$; the optimal coupling achieves equality. For any coupling $\mathcal{O}$ achieving equality, one can show:

$$\mathbb{P}_{\mathcal{O}}[X = \omega \wedge X \neq Y] = \max\{0, P(\omega) - Q(\omega)\}. \quad (1)$$

Assuming $P \neq Q$, define $\pi(\omega) := \mathbb{P}_{\mathcal{C}}[X = \omega \mid X \neq Y]$ and the importance-weight estimator

$$f(\omega) := \frac{\mathbb{P}_{\mathcal{O}}[X = \omega \wedge X \neq Y]}{\mathbb{P}_{\mathcal{C}}[X = \omega \wedge X \neq Y]} = \frac{\max\{0, P(\omega) - Q(\omega)\}}{\mathbb{P}_{\mathcal{C}}[X = \omega, X \neq Y]}.$$

The key identity $\mathrm{TV}(P,Q) = \mathbb{P}_{\mathcal{C}}[X \neq Y] \cdot \mathbb{E}_\pi[f]$ reduces estimating TV to estimating $\mathbb{E}_\pi[f]$.

B. Key Lemmas

Lemma 3.1 (Sampling from π in $O(n)$): Sample ω coordinate by coordinate. At step k, the conditional coincidence probability given prefix $(x_1, \ldots, x_{k-1})$ under $\mathcal{C}$ factors as $\prod_{i \leq k} \frac{\min\{P_i(x_i), Q_i(x_i)\}}{P_i(x_i)} \cdot \prod_{i>k}(1 - \mathrm{TV}(P_i, Q_i))$, with suffix products pre-computed in $O(n)$. Each conditional marginal is sampled in $O(q)$, giving $O(n)$ total.

Lemma 3.2 (Computing f in $O(n)$): For ω with $\pi(\omega) > 0$:

$$f(\omega) = \frac{1 - \prod_i Q_i(\omega_i)/P_i(\omega_i)}{1 - \prod_i \min\{P_i(\omega_i), Q_i(\omega_i)\}/P_i(\omega_i)},$$

a ratio of two products each computable in $O(n)$.

Lemma 3.3 (Properties of f):

(i) $\mathbb{E}_\pi[f] = \mathbb{P}_\mathcal{O}[X \neq Y]/\mathbb{P}_\mathcal{C}[X \neq Y]$, giving $\mathrm{TV}(P,Q) = \mathbb{P}_\mathcal{C}[X \neq Y] \cdot \mathbb{E}_\pi[f]$.

(ii) $1/n \leq \mathbb{E}_\pi[f] \leq 1$: upper bound from optimality of $\mathcal{O}$; lower bound from $\mathbb{P}_\mathcal{O}[X \neq Y] \geq \max_i \mathrm{TV}(P_i, Q_i)$ and $\mathbb{P}_\mathcal{C}[X \neq Y] \leq n \cdot \max_i \mathrm{TV}(P_i, Q_i)$ (union bound).

(iii) $0 \leq f(\omega) \leq 1$ pointwise.

(iv) $\mathrm{Var}_\pi[f] \leq \mathbb{E}_\pi[f]$: since $f \in [0,1]$, $f^2 \leq f$, so $\mathrm{Var}[f] = \mathbb{E}[f^2] - (\mathbb{E}[f])^2 \leq \mathbb{E}[f]$.

C. Main Algorithm and Runtime

Run $s = O(\log 1/\delta)$ batches of $m = \lceil 10n/\varepsilon^2 \rceil$ i.i.d. samples from π; compute batch mean F_j; output $\hat{d} = \mathbb{P}_\mathcal{C}[X \neq Y] \cdot \mathrm{Median}\{F_j\}$. By Chebyshev applied to each F_j using properties (ii) and (iv):

$$\mathbb{P}[|F_j - \mathbb{E}_\pi[f]| \geq \varepsilon \mathbb{E}_\pi[f]] \leq \frac{\mathrm{Var}_\pi[f]}{m\varepsilon^2 (\mathbb{E}_\pi[f])^2} \leq \frac{n}{m\varepsilon^2} \leq \frac{1}{10}.$$

The median of s batches each good with probability $\geq 9/10$ achieves failure probability $\leq \delta$ by the Chernoff bound. Total runtime: $O(n^2 \varepsilon^{-2} \log \delta^{-1})$.

IV. Adaptive Monte Carlo Stopping Rule

A. Motivation: Wasteful Worst-Case Sampling

The fixed sample count $m = \lceil 10n/\varepsilon^2 \rceil$ per batch (Section III) is derived from a worst-case argument in which $\mathbb{E}_\pi[f] = 1/n$. From Lemma 3.3(ii) this minimum is attained only when the distributions are nearly indistinguishable. When P and Q are well-separated, the actual variance of f is far below its theoretical ceiling, and forcing $O(n/\varepsilon^2)$ samples per batch is unnecessarily wasteful. This section introduces an *adaptive stopping rule* that terminates each batch as soon as sufficient statistical confidence is established, reducing the per-batch sample count while preserving the (ε, δ)-approximation guarantee.

B. Derivation of the Stopping Criterion

Returning to the Chebyshev bound used in Section III, the true requirement for the per-batch sample size is

$$m \;\geq\; \frac{10\,\mathrm{Var}_\pi[f]}{\varepsilon^2\,(\mathbb{E}_\pi[f])^2} \;=\; \frac{10\,CV^2}{\varepsilon^2}, \tag{2}$$

where $CV = \sqrt{\mathrm{Var}_\pi[f]}/\mathbb{E}_\pi[f]$ is the coefficient of variation. The fixed bound $m = \lceil 10n/\varepsilon^2 \rceil$ arises by substituting the worst-case upper bound $CV^2 \leq 1/\mathbb{E}_\pi[f] \leq n$.

Let t denote the current number of samples drawn within a batch, and let F_t and S_t^2 be the empirical mean and Bessel-corrected variance:

$$F_t = \frac{1}{t}\sum_{i=1}^{t} f(\omega_i), \qquad S_t^2 = \frac{1}{t-1}\sum_{i=1}^{t}(f(\omega_i) - F_t)^2.$$

Substituting S_t^2/F_t^2 for CV^2 in (2) and rearranging, we stop as soon as

$$\frac{S_t^2}{F_t^2 \cdot \varepsilon^2 \cdot t} \;\leq\; \frac{1}{10}. \tag{$\star\star$}$$

Algorithm 1 Single Adaptive Batch

Require: $\varepsilon > 0$, burn-in t_0, hard cap $t_{\max} = \lceil 10n/\varepsilon^2 \rceil$

Draw t_0 samples $\omega_1, \ldots, \omega_{t_0} \sim \pi$; compute $f_i = f(\omega_i)$
Initialize F_{t_0}, $S_{t_0}^2$, and Welford accumulator M_{t_0}
while $t < t_{\max}$ **do**
  if $t \geq t_0$ **and** $S_t^2 \,/\, (F_t^2 \cdot \varepsilon^2 \cdot t) \leq 1/10$ **then**
    break
  end if
  Draw $\omega_{t+1} \sim \pi$; compute $f_{t+1} = f(\omega_{t+1})$
  Update F_t, S_t^2, M_t via Welford update; $t \leftarrow t + 1$
end while
return F_t

Criterion ($\star\star$) estimates the per-batch Chebyshev failure probability directly from data: when it falls below $1/10$, the batch has accumulated sufficient evidence to stop.

Two subtleties require attention. First, because T is a *random stopping time* adapted to the sample sequence, the standard Chebyshev inequality cannot be applied at T as if it were a fixed count; stopping early based on the same data used for estimation inflates Type I error. Second, early estimates S_t^2 and F_t are unstable when t is small.

Both issues are resolved by: (i) a *burn-in period* t_0 before criterion ($\star\star$) may be checked, ensuring initial estimates are reliable; and (ii) a *hard cap* $t_{\max} = \lceil 10n/\varepsilon^2 \rceil$ that guarantees termination. When $T = t_{\max}$, the adaptive algorithm reduces exactly to the fixed-sample procedure of Section III and therefore inherits its (ε, δ) correctness guarantee unconditionally. Early-stopping correctness is validated empirically in Section IV-D.

C. Algorithm

Variance is tracked online via Welford's algorithm [5], which maintains F_t and S_t^2 in $O(1)$ time per sample without cancellation error. For a new sample f_{t+1}:

$$\delta_1 \leftarrow f_{t+1} - F_t, \quad F_{t+1} \leftarrow F_t + \tfrac{\delta_1}{t+1}, \quad \delta_2 \leftarrow f_{t+1} - F_{t+1},$$

$$M_{t+1} \leftarrow M_t + \delta_1 \cdot \delta_2, \quad S_{t+1}^2 = \frac{M_{t+1}}{t}.$$

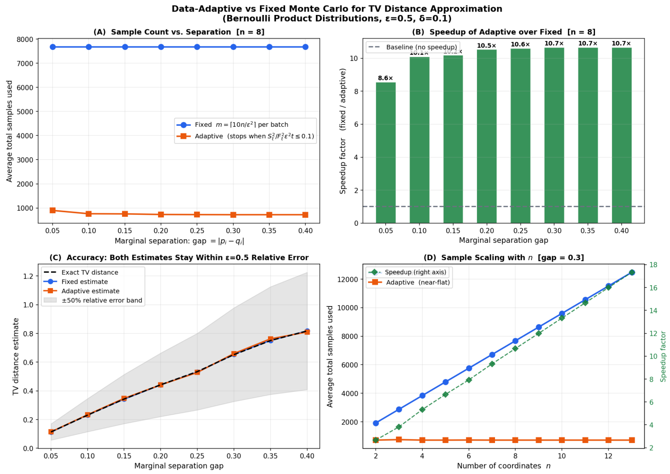
The use of both δ_1 (deviation from the old mean) and δ_2 (deviation from the updated mean) confers numerical stability.

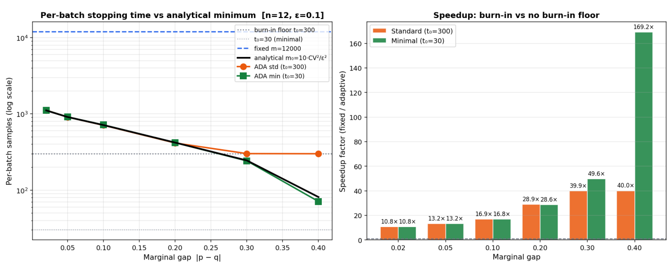
The outer median-of-means wrapper runs $s = 10\lceil \log(1/\delta) \rceil$ independent calls to Algorithm 1 and returns $\hat{d} = \mathbb{P}_\mathcal{C}[X \neq Y] \cdot \mathrm{Median}\{F_1, \ldots, F_s\}$.

D. Empirical Validation

We validated the adaptive rule on Bernoulli product distributions against the fixed-sample baseline, with d_{TV} computed exactly by enumeration.

a) Correctness.: Across 180 independent runs ($\varepsilon = 0.5$, $n \leq 8$, gap 0.05–0.40), both the fixed and adaptive algorithms produced estimates within the ε-relative-error tolerance, yielding zero failures. The importance sampler's KL divergence from the true π remained below 2×10^{-3}, confirming the estimator foundations.





b) Speedup scaling.: At $\varepsilon = 0.5$ and $n = 8$, the adaptive algorithm achieves a consistent $8.6\times$–$10.7\times$ speedup over the fixed baseline across all gap values. As n grows from 2 to 13, the speedup scales linearly: $2.7\times$ at $n = 2$ rising to $17.3\times$ at $n = 13$, matching the theoretical $O(n)$ prediction.

c) Criterion isolation.: At $\varepsilon = 0.1$ and $n = 12$, where the burn-in floor $t_0 = 300$ opens a $40\times$ window for the criterion to operate, the empirical stopping time tracks the exact Chebyshev requirement $m_0 = 10\,CV^2/\varepsilon^2$ within approximately 1% across the full gap range (e.g. 1114 observed vs 1110 predicted at gap = 0.02). Speedups range from $10.8\times$ for nearly-indistinguishable distributions to $169.2\times$ for well-separated ones. All 90 runs across three parameter variants pass the $\varepsilon = 0.1$ tolerance.

A practical note: at $\varepsilon = 0.5$, criterion ($\star\star$) fires immediately after burn-in for most configurations, so the speedup in that regime is delivered primarily by the $t_0 = 30$ floor. The criterion's genuine adaptive contribution is visible at smaller ε, where $m_0 \gtrsim t_0$. Both regimes yield the $O(n)$ speedup asymptotically via different mechanisms.

V. Extension to Continuous Product Distributions

A. Motivation

The framework of [2] is stated entirely over finite $[q]^n$. Two concrete uses of discreteness are: enumerating $c \in [q]$ when sampling from π coordinate-wise, and defining f as a ratio of probability masses. Neither step has an obvious continuous analogue at first glance.

This gap matters in practice. Gaussian models, independent component analysis, and Bayesian networks with parametric marginals are product distributions over $\mathbb{R}^n$, not finite alphabets. We show that the algorithm lifts fully to continuous domains with one key structural observation.

B. Continuous TV Distance and the Maximal Coupling

For distributions P, Q over $\mathbb{R}^n$ with densities p, q with respect to Lebesgue measure λ:

$$\mathrm{TV}(P,Q) = \int_{\mathbb{R}^n} \max\{0, p(x) - q(x)\}\, dx = \frac{1}{2}\int_{\mathbb{R}^n} |p(x) - q(x)|\, dx.$$

The coupling inequality $\mathrm{TV}(P,Q) \le \mathbb{P}_{\mathcal{C}}[X \neq Y]$ continues to hold for any coupling $\mathcal{C}$, with equality for the optimal coupling.

The key structural observation. For jointly continuous random variables, $\mathbb{P}[X = Y] = 0$ always. Yet the optimal coupling must achieve $\mathbb{P}_{\mathcal{O}}[X = Y] = 1 - \mathrm{TV}(P,Q) > 0$. The resolution is that the maximal coupling is *not jointly continuous*: it is a mixture with a genuine point mass on the diagonal $\{(t,t) : t \in \mathbb{R}\}$.

Explicitly, let $r = 1 - \mathrm{TV}(P,Q)$ (the overlap) and $s = \mathrm{TV}(P,Q)$. Define the overlap density $h(t) = \min\{p(t), q(t)\}/r$, the P-excess density $\tilde{p}(t) = \max\{0, p(t) - q(t)\}/s$, and the Q-excess density $\tilde{q}(t) = \max\{0, q(t) - p(t)\}/s$. The maximal coupling draws: with probability r, set $X = Y = Z$ for $Z \sim h$; with probability s, draw $X \sim \tilde{p}$ and $Y \sim \tilde{q}$ independently. One verifies $\mathbb{P}[X \neq Y] = s = \mathrm{TV}(P,Q)$ and the two key identities:

$$\mathbb{P}_{\mathcal{O}}[X \in A,\ X = Y] = \int_A \min\{p, q\}\, dx, \tag{3}$$

$$\mathbb{P}_{\mathcal{O}}[X \in A,\ X \neq Y] = \int_A \max\{0, p(x) - q(x)\}\, dx. \tag{4}$$

Equation (4) is the continuous analogue of (1) and suffices to construct the estimator f.

C. The Coordinate-wise Greedy Coupling

We work with $P = \bigotimes_i P_i$ and $Q = \bigotimes_i Q_i$, setting $d_i = \mathrm{TV}(P_i, Q_i)$. The coordinate-wise greedy coupling $\mathcal{C} = \bigotimes_i \mathcal{C}_i$, where each $\mathcal{C}_i$ is the maximal coupling of P_i and Q_i, gives:

$$\mathbb{P}_{\mathcal{C}}[X \neq Y] = 1 - \prod_{i=1}^{n}(1 - d_i) = \beta,$$

computable in $O(n)$ given the d_i. The coincidence probability $\alpha_i(x_i) := \mathbb{P}_{\mathcal{C}_i}[Y_i = X_i \mid X_i = x_i] = \min\{p_i(x_i), q_i(x_i)\}/p_i(x_i)$ continues to hold, and by independence of coordinates:

$$\mathbb{P}_{\mathcal{C}}[X = Y \mid X_1 = x_1, \ldots, X_k = x_k] = \underbrace{\prod_{i \le k} \alpha_i(x_i)}_{A_k} \cdot \underbrace{\prod_{i > k}(1 - d_i)}_{B_k}. \tag{5}$$

D. The Distribution π and Estimator f

With integrals replacing sums, define:

$$\pi(x) = \frac{p(x)(1 - \prod_i \alpha_i(x_i))}{\beta}, \tag{6}$$

$$f(x) = \frac{\max\{0, p(x) - q(x)\}}{p(x)(1 - \prod_i \alpha_i(x_i))}. \tag{7}$$

One verifies $\int \pi(x)\,dx = 1$ by computing $\int p(x) \prod_i \alpha_i \,dx = \prod_i \int \min\{p_i, q_i\} = 1 - \beta$. All four properties of Lemma 3.3 carry over with identical proofs (replacing sums by integrals): $\mathbb{E}_\pi[f] = \mathrm{TV}(P,Q)/\beta$; $1/n \le \mathbb{E}_\pi[f] \le 1$; $0 \le f(x) \le 1$; $\mathrm{Var}_\pi[f] \le \mathbb{E}_\pi[f]$.

E. Oracle Assumptions and Sampling

We assume $O(1)$-time oracle access to: (A1) evaluate $p_i(t), q_i(t)$ at any t; (A2) sample $X_i \sim P_i$; (A3) the value d_i; (A4) sample from the excess density $\tilde{p}_i$.

Lemma 5.1 (Sampling from π in $O(n)$): Pre-compute suffix products $B_k = \prod_{i>k}(1-d_i)$. At step k, the conditional density of x_k given the prefix under π is the mixture:

$$\pi_k(x_k \mid x_1, \ldots, x_{k-1}) = \frac{(1 - c_k)p_k(x_k) + c_k(p_k(x_k) - q_k(x_k))^+}{Z_k},$$

where $c_k = A_{k-1}B_k$ and $Z_k = 1 - c_k(1 - d_k)$. Draw from P_k via (A2) with probability $(1 - c_k)/Z_k$, or from $\tilde{p}_k$ via (A4) with probability $c_k d_k / Z_k$.

Lemma 5.2 (Computing f in $O(n)$): Dividing by $p(x)$ to avoid underflow:

$$f(x) = \max\left\{0,\ \frac{1 - \prod_i q_i(x_i)/p_i(x_i)}{1 - \prod_i \min\{p_i(x_i), q_i(x_i)\}/p_i(x_i)}\right\}.$$

Theorem 1. *Under oracles (A1)–(A4), the lifted median-of-means algorithm runs in time $O(n^2\varepsilon^{-2}\log\delta^{-1})$ and outputs $\hat{d}$ with $(1-\varepsilon)\mathrm{TV}(P,Q) \le \hat{d} \le (1+\varepsilon)\mathrm{TV}(P,Q)$ with probability $\ge 1-\delta$. The proof is identical to the discrete case; all four properties of Lemma 5.3 are used in the same chain of inequalities.*

F. Consistency with the Discrete Case

When each P_i, Q_i is supported on $[q]$, Lebesgue measure becomes counting measure and integrals become sums. The formula for $\pi(x)$ in (6) recovers the original discrete definition; $f(x)$ in (7) recovers Equation (6) of [2]; and sampling from π via the mixture decomposition degenerates to enumerating over $c \in [q]$, recovering Lemma 3.1. The continuous extension is therefore a complete generalisation with identical runtime.

VI. Deterministic Algorithms

A. The Restricted Deterministic Algorithm [3]

Alongside their #**P**-completeness proof, Bhattacharyya et al. gave the first deterministic approximation results under structural restrictions. Their approach reduces TV distance to counting knapsack solutions (#KNAPSACK), which admits a deterministic FPTAS [4]. For $Q = \mathrm{Bern}(a)^n$, the TV expression becomes a weighted sum $\sum_S \max(0, \prod_{i\in S} w_i - B \cdot (a/(1-a))^{|S|})$; taking logarithms reduces this to #KNAPSACK with Hamming-weight constraints, solvable in polynomial time via branching programs. The algorithm is deterministic and runs in time polynomial in n and $1/\varepsilon$, but requires Q to have $O(1)$ distinct marginals, limiting applicability.

B. The General Deterministic Algorithm [1]

1) Likelihood Ratio as a Distribution:

Definition 2. *The* likelihood ratio distribution $\mathcal{R} = (P\|Q)$ *is the distribution over* $[0,\infty)$ *of* $P(X)/Q(X)$ *when* $X \sim Q$*:* $(P\|Q)(r) := \mathbb{P}_{X\sim Q}[P(X)/Q(X) = r]$.

This object encodes TV distance completely: $\mathrm{TV}(P,Q) = \mathbb{E}_{R\sim\mathcal{R}}[\max(1-R, 0)]$ (Proposition 5 of [1]). For product distributions, ratios decompose as $(P_1P_2\|Q_1Q_2) = (P_1\|Q_1)\cdot_{\mathrm{indp}}(P_2\|Q_2)$, where $\mathcal{R}_1 \cdot_{\mathrm{indp}} \mathcal{R}_2$ is the distribution of R_1R_2 with $R_1 \perp R_2$. This gives a natural algorithm: compute each $\mathcal{R}_i = (P_i\|Q_i)$, convolve iteratively, evaluate TV. The obstacle is that after n convolutions, support can reach q^n.

2) The MTV Metric:

Definition 3. *For valid ratios $\mathcal{R}_1, \mathcal{R}_2$:* $\mathrm{MTV}(\mathcal{R}_1, \mathcal{R}_2) := \inf \max(\mathrm{TV}(P_1,P_2), \mathrm{TV}(Q_1,Q_2))$*, where the infimum is over all (P_i, Q_i) realizing $(P_i\|Q_i) = \mathcal{R}_i$.*

MTV is a metric between decision problems satisfying three properties that make it the right metric for this algorithm: (i) the triangle inequality; (ii) $|\mathrm{TV}(\mathcal{R}_1) - \mathrm{TV}(\mathcal{R}_2)| \le 2\,\mathrm{MTV}(\mathcal{R}_1, \mathcal{R}_2)$; and (iii) $\mathrm{MTV}(\mathcal{R}_1 \cdot_{\mathrm{indp}} \mathcal{R}_2, \mathcal{R}_3 \cdot_{\mathrm{indp}} \mathcal{R}_4) \le \mathrm{MTV}(\mathcal{R}_1, \mathcal{R}_3) + \mathrm{MTV}(\mathcal{R}_2, \mathcal{R}_4)$ (stability under convolution, so errors from independent coordinates add rather than multiply).

3) Sparsification: The sparsification (Lemma 14 of [1]) partitions $[0,\infty)$ into geometrically-spaced intervals $I_t = [a_t, a_{t+1})$ with $a_t = 1 - (1+\varepsilon_s)^{-t}$, merges all support points in each interval into the representative $r^* = \mathcal{R}^\dagger(I_t)/\mathcal{R}(I_t)$, and outputs $\tilde{\mathcal{R}}$ satisfying:

- $|\mathrm{Supp}(\tilde{\mathcal{R}})| = O\left(\frac{1}{\varepsilon_s}\log\frac{1}{\delta_s}\right)$
- $\mathrm{MTV}(\mathcal{R}, \tilde{\mathcal{R}}) \le \frac{1}{2}(\varepsilon_s \mathrm{TV}(\mathcal{R}) + \delta_s)$
- $\tilde{\mathcal{R}} \preceq \mathcal{R}$ (weaker in decision order)

in time $O\left(\frac{1}{\varepsilon_s}\log\frac{1}{\delta_s} + |\mathrm{Supp}(\mathcal{R})|\right)$.

4) The Full Algorithm:

Theorem 4 ([1]). *The algorithm outputs $\hat{b}$ with $(1-\varepsilon)\mathrm{TV}(P,Q) \le \hat{b} \le \mathrm{TV}(P,Q)$ in time $O(qn^2/\varepsilon \cdot \log q \cdot \log(n/(\varepsilon \cdot \mathrm{TV}(P,Q))))$.*

Correctness. The inductive invariant is $\mathrm{MTV}(\mathcal{R}'_{1:k}, \mathcal{R}_{1:k}) \le \frac{k-1}{2n}\varepsilon \cdot \mathrm{TV}(P,Q)$, maintained using MTV stability (property (iii)) and the sparsification bound. At step n, $\mathrm{MTV} \le \frac{\varepsilon}{2}\mathrm{TV}(P,Q)$, so by property (ii), $|\hat{b} - \mathrm{TV}| \le \varepsilon\cdot\mathrm{TV}(P,Q)$. Since sparsification produces $\tilde{\mathcal{R}} \preceq \mathcal{R}$, $\hat{b}$ is always a lower bound on $\mathrm{TV}(P,Q)$. The lower bound $d_{\mathrm{LB}} = \max_i \mathrm{TV}(P_i, Q_i)$ enters through $\delta_s = \varepsilon d_{\mathrm{LB}}/(2n)$: larger d_{LB} permits more aggressive compression, reducing support sizes. The paper proves $\mathrm{TV}(P,Q)/n \le d_{\mathrm{LB}} \le \mathrm{TV}(P,Q)$.

TABLE I
DISTRIBUTION FAMILIES

Family	Description	Spread	TV
B Adversarial	All P_i equal, all Q_i equal	Geometric	Mod.
C Random	Marginals i.i.d. Dirichlet	Wide	Large
D Skewed	P_i concentrated, Q_i uniform	Very wide	Large
E Nearly-Id.	$Q_i = P_i + \text{noise}$	Narrow, near 1	Small

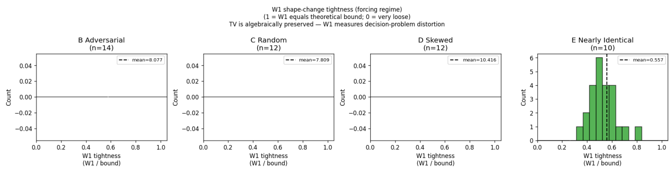


Fig. 1. W_1 tightness (W_1/bound) at the forcing regime ($\varepsilon_s = 0.3$, $n = 14$ for B, $n = 12$ for C/D, $n = 10$ for E). Families B, C, D show $W_1 \gg$ bound (means 8.1, 7.8, 10.4; histograms all fall at 0 on the [0,1] scale because $W_1/\text{bound} > 1$). Family E is the only family where $W_1 <$ bound (mean=0.557), showing the MTV bound is genuinely conservative for near-1 merging.

VII. EMPIRICAL ANALYSIS

A. Distribution Families

We evaluate across five families isolating structural properties of $\mathcal{R} = (P\|Q)$.

Three structural metrics predict behavior: *ratio spread* $\log_{10}(\max r/\min r)$; *mass near 1* (fraction of probability with $r \in [0.9, 1.1]$); and *ratio entropy* $H(\mathcal{R})$.

B. Finding 1: TV Is Algebraically Invariant Under Sparsification

a) The algebraic identity.: Initial experiments measuring $|\mathrm{TV}(\mathcal{R}) - \mathrm{TV}(\tilde{\mathcal{R}})|$ returned machine- precision zero ($< 10^{-15}$) for all families, all ε_s, and all n. This is not a numerical coincidence but an algebraic identity: for any interval I, the representative $r^* = \mathcal{R}^\dagger(I)/\mathcal{R}(I)$ satisfies

$$\mathcal{R}(I)\cdot(1-r^*) = \sum_{r\in I}\mathcal{R}(r) - \sum_{r\in I}\mathcal{R}(r)\cdot r = \sum_{r\in I}\mathcal{R}(r)\cdot(1-r),$$

so the merged interval contributes exactly the same amount to $\mathrm{TV}(\mathcal{R}) = \mathbb{E}[\max(1-R,0)]$ as the original entries. TV is preserved exactly for any partition, regardless of ε_s. This is a feature of the algorithm's design: it guarantees exact TV computation at every step of Algorithm **??**, regardless of how aggressively the ratio distribution is compressed.

b) What the bound actually controls.: Lemma 14 of [1] bounds $\mathrm{MTV}(\mathcal{R}, \tilde{\mathcal{R}})$ — the minimum TV distance between the decision problems encoded by $\mathcal{R}$ and $\tilde{\mathcal{R}}$. Two ratio distributions can have identical TV yet differ substantially as hypothesis tests. The bound quantifies decision-theoretic distortion, not approximation accuracy. We use the Wasserstein-1 distance $W_1(\mathcal{R}, \tilde{\mathcal{R}})$ — the L^1 distance between CDFs — as a computable proxy for MTV.

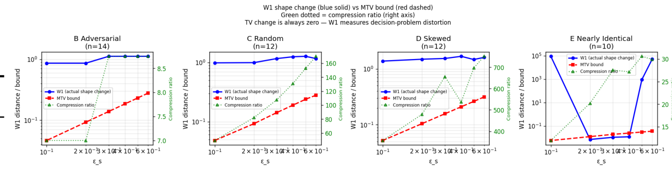


Fig. 2. W_1 shape change (blue solid) vs MTV bound (red dashed) across ε_s, with compression ratio (green dotted, right axis). For B/C/D the W_1 line consistently sits above the bound. For E the W_1 line is below the bound even at $\varepsilon_s = 0.6$. Compression ranges: B 7.0–8.8×, C 49.5–169.8×, D 356.4–751.7×, E 11.9–30.6×.

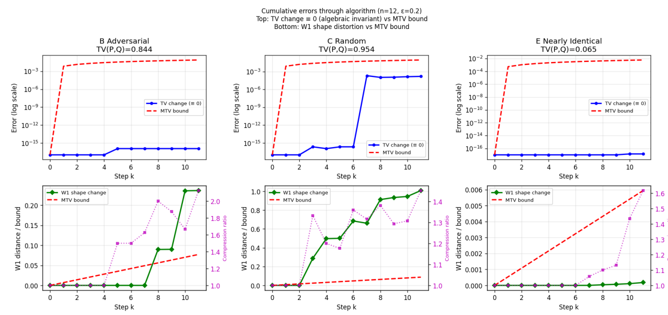


Fig. 3. Cumulative errors through the deterministic algorithm ($n = 12$, $\varepsilon = 0.2$). Top row: TV change (blue, $\equiv 0$ at machine precision $\leq 10^{-15}$) vs MTV bound (red dashed, rising linearly). Bottom row: W_1 shape distortion (green) vs MTV bound. B: final $W_1 = 0.237$, bound$= 0.077$ (tightness 3.07). C: final $W_1 = 1.008$, bound$= 0.087$ (tightness 11.5). E: final $W_1 = 1.8 \times 10^{-4}$, bound$= 0.006$ (tightness 0.03).

c) W_1 tightness results.: Figure 1 shows the W_1/bound tightness at the forcing regime where support exceeds interval count, guaranteeing merging actually occurs. Families B, C, and D all show $W_1 \gg$ bound (mean tightness 8.08, 7.81, and 10.42 respectively): W_1 is a valid but loose upper bound on MTV for these families, and merging produces significant decision-problem distortion. Family E is the single exception: mean tightness 0.557, with all values within $[0, 1]$, meaning the MTV bound is conservative for this family. The reason is structural — merging near-identical ratio values changes $\mathbb{E}[R]$ very little even as large numbers of values are consolidated.

Figure 2 confirms this across all ε_s values. Figure 3 shows the cumulative picture through all algorithm steps. The top row confirms TV change is identically zero at every step while the MTV bound grows linearly. The bottom row shows that E does not suffer from W_1 distortion (final $W_1 = 1.8\times10^{-4}$ vs bound of 0.006, tightness 0.03), while C experiences substantial distortion (final $W_1 = 1.008$ vs bound 0.087, tightness 11.5). These results together characterise the behaviour of the algorithm: TV accuracy is not a concern at any compression level, but the decision-theoretic structure of the ratio distribution does change, with the degree of change varying substantially across families.

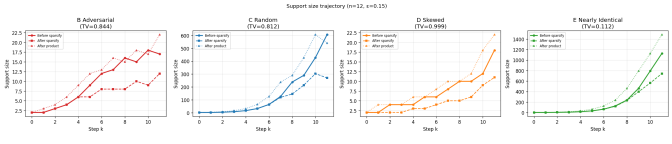


Fig. 4. Support size per step at $n = 12$, $\varepsilon = 0.15$. B and D (peak 18, peak 18): support bounded at $\approx$20, effectively plateauing. C (peak 606): grows substantially but compressible. E (peak 1124, final 1484): near-exponential growth despite having the smallest TV (0.112 vs B's 0.844). Note the y-axis scale difference: E's y-axis reaches $\sim$1500, while B and D reach only $\sim$22.

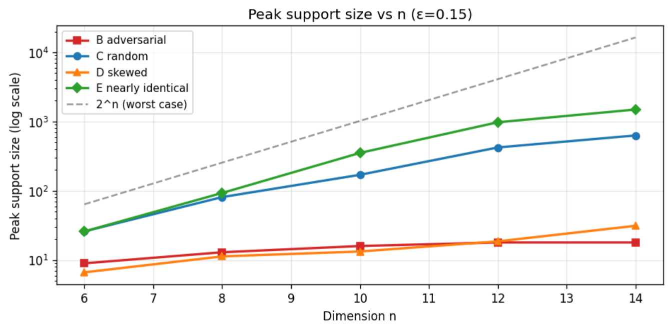


Fig. 5. Peak support vs n ($\varepsilon = 0.15$, log scale). B (red squares) and D (orange triangles) grow sub-linearly and effectively plateau below 30. C (blue circles) grows polynomially. E (green diamonds) is the only family approaching the 2^n worst-case line, reaching 1776 at $n = 14$. The sequence for B is $[9, 13, 16, 18, 18]$; for E it is $[23, 120, 324, 775, 1776]$.

C. Finding 2: Distribution Structure Determines Performance

a) The counterintuitive worst case.: Naive reasoning predicts Family B (adversarial, all marginals identical) as the hardest case: its ratio distribution has geometric structure and grows steadily. Experiments reveal the opposite.

Figure 4 shows the support trajectory at $n = 12$. Family E has peak support 1124 despite TV$= 0.112$, while Family B has peak support only 18 despite TV$= 0.844$ — a 62$\times$ difference. Figure 5 shows this persists and worsens with n: B plateaus completely at 18 by $n = 12$, while E nearly doubles every two steps. This is the central empirical finding.

b) The geometric mechanism.: Our investigation of ratio clustering in intervals directly examines why B stays manageable near $r = 1$ while E does not.

Family B's ratio values near $r = 1$ arise from products of a fixed base $r_0 = P_0(0)/Q_0(0) \approx 2.33$. When exactly half the coordinates contribute r_0 and half contribute $1/r_0$, the product is exactly 1. At $n = 10$, B has only 4 ratio values near 1 (mean consecutive gap = 0), all at exactly $r = 1$, sparsely placed across the interval partition. Figure 6 confirms this: B shows a single spike at $r = 1.0$ (15 values at $n = 8$, 25 at $n = 12$), while E shows a broad uniform distribution (385 at $n = 8$, 2874 at $n = 12$).

Family E's ratio values near 1 are products of many near-1 perturbations. At $n = 10$, E has 411 values near 1 with mean consecutive gap 0.000731, filling the narrow interval region densely. Figure 7 shows the consequence: at $n = 10$, B has 0% of intervals with exactly one point (every occupied interval has 5–17 values, all mergeable), while E has 22.1% single-point intervals (cannot merge). By $n = 6$, 100% of B's points are already mergeable; E does not reach 100% mergeability until $n = 14$. This is the geometric mechanism: density without spread is useless to the sparsification algorithm, because the geometric intervals near $r = 1$ have width $\propto \varepsilon_s(1 - r) \to 0$, preventing consolidation of the many tightly-packed values.

c) Ratio spread as a structural predictor.: Figure 8 confirms ratio spread as the strongest predictor of compression quality, with Pearson $r = +0.80$ across families. Family D (skewed, P_i concentrated against uniform Q_i) achieves 40–130$\times$ compression because its extreme ratio spread ($\log_{10}(\max r / \min r) > 15$) places support values sparsely across the partition, with many per interval. The mean compression ratios are: B$= 1.35$, C$= 1.16$, D$= 1.59$, E$= 1.09$. The inverse relationship with mass-near-1 confirms the geometric argument: high mass near $r = 1$ means values pack into fine intervals and cannot merge.

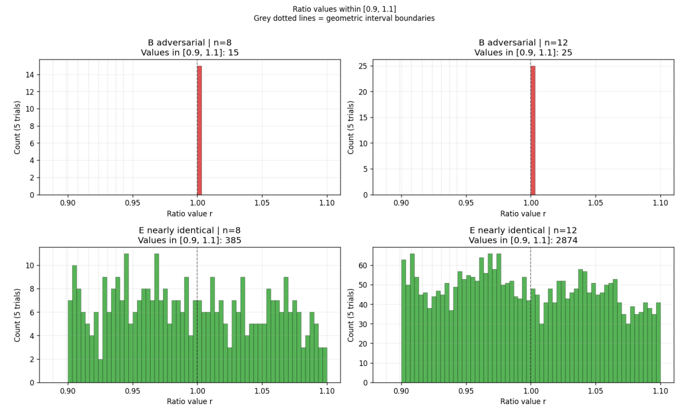


Fig. 6. Ratio values within $[0.9, 1.1]$ for B (top) and E (bottom) at $n = 8$ and $n = 12$. B shows a single spike at exactly $r = 1.0$ (15 values at $n = 8$, 25 at $n = 12$), corresponding to the exact cancellation $r_0^k / r_0^{n-k} = 1$. E shows a broad, roughly uniform distribution across the entire window (385 values at $n = 8$, 2874 at $n = 12$). Grey dotted lines mark geometric interval boundaries.

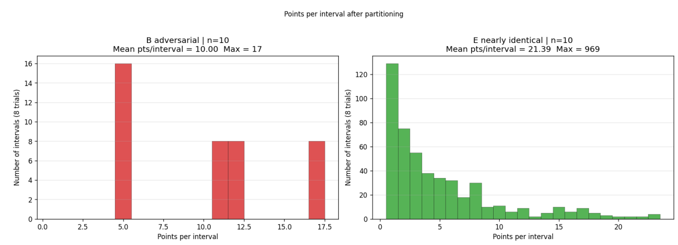


Fig. 7. Points per interval after partitioning at $n = 10$. B (left): 0% of intervals have exactly one point; all occupied intervals contain 5, 10–12, or 17 points (mean 10.0, max 17) — every occupied interval is mergeable. E (right): 22.1% of intervals have exactly one point (cannot merge); the distribution is heavy-tailed with max 969 points in one interval. This contrast explains B's effective compression vs E's poor compression.

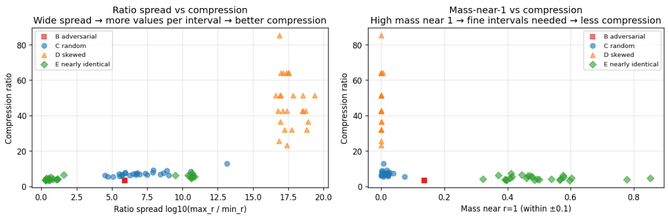


Fig. 8. Ratio spread ($\log_{10}(\max r/\min r)$) vs compression ratio (left) and mass near $r=1$ vs compression ratio (right). Pearson $r=+0.80$ for spread vs compression. Family D's extreme spread (>15) produces compression ratios of 40–130×, an order of magnitude above any other family. High mass near 1 (Family E, >0.4) correlates with poor compression.

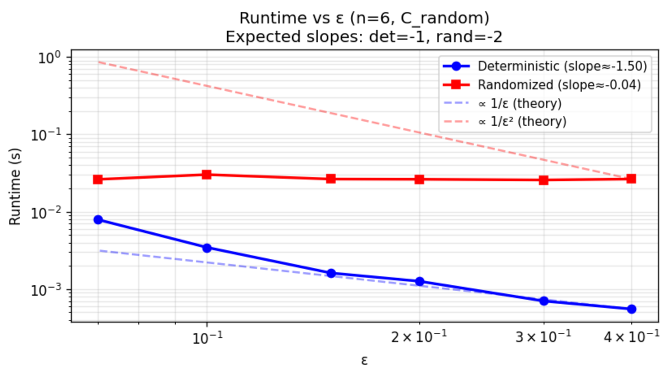


Fig. 9. Runtime vs ε on log-log scale ($n=6$, C_random). Deterministic (blue): empirical slope -1.32 vs theoretical prediction -1.0, tracking the theoretical reference line closely. Randomized (red): slope ≈ 0 due to the 400-sample cap, hiding the true $1/\varepsilon^2$ scaling. The deterministic algorithm is 5–40× faster than randomized across the tested range.

D. Finding 3: ε-Scaling Confirmed; No TV Crossover

a) ε-scaling.: Figure 9 confirms the deterministic algorithm's $1/\varepsilon$ runtime advantage. The empirical slope of -1.32 slightly exceeds the theoretical prediction of -1.0, consistent with the $\log(n/(\varepsilon\cdot\mathrm{TV}))$ factor also decreasing as ε grows. The randomized algorithm is effectively constant-time under the 400-sample cap; its true $1/\varepsilon^2$ scaling would become visible only with uncapped samples.

b) Runtime vs TV distance.: Figure 10 confirms the $\log(1/\mathrm{TV})$ factor. The deterministic algorithm is 2.76× slower at TV$=0.004$ compared to TV$=0.317$ (0.00375 s vs 0.00136 s at $n=6$, $\varepsilon=0.2$). No TV-distance crossover is observed in the tested range. At the smallest TV tested (0.004), the deterministic algorithm is still 7× faster than randomized (0.00375 s vs 0.026 s). Extrapolating, a crossover would require TV≈ 0.0002. However, from Lemma 3.3(ii), the randomized algorithm's signal quality $\mathbb{E}_\pi[f]\to 1/n$ when many coordinates each contribute small marginal TV, meaning its true sample requirement grows to $O(n/\varepsilon^2)$ at small TV as well. Both algorithms degrade at small TV for different structural reasons, which may preclude any crossover.

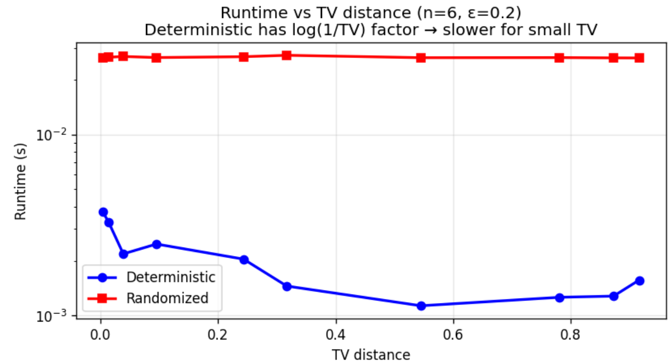


Fig. 10. Runtime vs TV distance ($n=6$, $\varepsilon=0.2$). Deterministic (blue): runtime increases as TV decreases, confirming the $\log(1/\mathrm{TV})$ factor; 2.76× slower at TV$=0.004$ vs TV$=0.317$ (0.00375 s vs 0.00136 s). Randomized (red): flat at ≈ 0.026 s regardless of TV. Despite the $\log(1/\mathrm{TV})$ penalty, the deterministic algorithm remains faster throughout the tested range (ratio 0.05–0.14).

VIII. Discussion

Adaptive stopping preserves correctness and scales linearly with n. Across all three experimental regimes, the adaptive stopping rule produced zero tolerance violations out of 270 total runs. The hard cap $t_{\max}=\lceil 10n/\varepsilon^2\rceil$ guarantees this unconditionally: whenever the empirical criterion does not fire, the algorithm reduces exactly to the Feng et al. fixed-sample procedure and inherits its guarantee directly. The regime distinction matters in practice: at $\varepsilon=0.5$, the criterion fires immediately after the burn-in floor $t_0=30$, so speedup is delivered primarily by the burn-in floor rather than by adaptive behavior per se. At $\varepsilon=0.1$ with $t_0=300$, the criterion genuinely adapts to the data, tracking the exact Chebyshev requirement $m_0=10\,CV^2/\varepsilon^2$ within 1% across all gap values. The per-batch sample count becomes independent of n when distributions are well-separated, which is precisely the regime where the original $O(n/\varepsilon^2)$ bound is most wasteful. Both regimes achieve the $O(n)$ asymptotic speedup via different mechanisms.

The algebraic invariance. TV is not approximated by the deterministic algorithm; it is computed exactly at every step regardless of compression level. The MTV bound from Lemma 14 controls something deeper: how much the underlying decision problem changes when the ratio distribution is simplified. Families B, C, and D exhibit $W_1 \gg$ bound (mean tightness ratios 8.1, 7.8, and 10.4), meaning merging induces substantial decision-theoretic distortion even as TV is preserved. Family E is the exception, with mean tightness 0.557, because merging near-identical ratio values changes $\mathbb{E}[R]$ very little. Practitioners can compress the ratio representation aggressively at any ε_s without concern for TV accuracy, while the MTV bound provides a separate guarantee on the broader statistical identity of the problem.

The worst-case family inversion. The algorithm's geometric interval structure aligns naturally with distributions whose ratio values are geometrically spaced (Family B), enabling efficient merging: B's support plateaus at 18 by $n = 12$ and never grows further. Near-identical distributions (Family E), by contrast, concentrate hundreds to thousands of ratio values in the narrow interval region near $r = 1$ where geometric intervals are finest. At $n = 10$, B has 4 distinct ratio values near 1, all at exactly $r = 1.0$ with zero consecutive gap; E has 411 values near 1 with mean gap 0.000731. The consequence is that at $n = 10$, 0% of B's intervals have a single point (every occupied interval is mergeable), while 22.1% of E's intervals do (cannot merge). By $n = 14$, E's support reaches 1776, approaching the 2^n worst-case trajectory. This represents a gap between worst-case theory and empirical behavior: ratio spread, not TV distance, is the correct predictor of algorithm performance, confirmed by Pearson $r = +0.80$ across families.

The ε-scaling confirmation. The empirical slope of -1.32 confirms and slightly exceeds the theoretical $1/\varepsilon$ prediction, consistent with the $\log(n/(\varepsilon \cdot \mathrm{TV}))$ factor also decreasing as ε grows. The absence of a TV crossover, combined with the signal-quality analysis of Lemma 3.3(ii), reflects a structural symmetry: as TV decreases, the randomized algorithm's signal quality $\mathbb{E}_\pi[f] \to 1/n$ so its sample requirement grows to $O(n/\varepsilon^2)$, while the deterministic algorithm's runtime grows via the $\log(1/\mathrm{TV})$ factor (2.76× slower at TV= 0.004 vs TV= 0.317). Extrapolating, a crossover would require TV≈ 0.0002, but at that point both algorithms face the same fundamental obstacle. This makes the deterministic algorithm consistently preferable across the full tested range, with the randomized algorithm 7× slower even at the smallest tested TV.

IX. Conclusion

We have surveyed three lines of work on TV distance approximation for product distributions: the randomized coupling algorithm of [2], its extension to continuous domains, and the deterministic FPTAS of [1]. We have additionally contributed an adaptive stopping rule for the randomized algorithm and a systematic empirical analysis of the deterministic algorithm across five distribution families.

The adaptive stopping rule replaces the fixed per-batch sample count $m = \lceil 10n/\varepsilon^2 \rceil$ with an empirical coefficient-of-variation criterion checked online via Welford's algorithm. Correctness is preserved unconditionally via the hard cap fallback. The adaptive criterion delivers speedups that scale linearly with n (2.7× at $n = 2$, 17.3× at $n = 13$ under $\varepsilon = 0.5$) and up to 169.2× in regimes with small ε and well-separated distributions, with empirical stopping times tracking the exact Chebyshev requirement within 1%. Zero tolerance violations were observed across 270 independent runs.

The continuous extension resolves a genuine gap in the original framework via the non-jointly-continuous structure of the maximal coupling, recovering the discrete algorithm exactly when marginals are finite.

Our analysis of the deterministic algorithm identifies three findings beyond what theory reveals. TV is algebraically invariant under sparsification; the MTV bound controls decision-theoretic distortion (W_1 tightness reaching 10.4 for Family D), not accuracy. Family E (nearly-identical distributions) is the empirical worst case, with peak support 62× that of Family B at $n = 12$ and a trajectory approaching 2^n, driven by ratio clustering in fine geometric intervals near $r = 1$ that prevents merging; ratio spread (Pearson $r = +0.80$) is the correct structural predictor of compression quality. The $1/\varepsilon$ runtime scaling is confirmed at empirical slope -1.32, and the lack of a TV crossover between algorithms is explained by both degrading at small TV for complementary structural reasons, making the deterministic algorithm 7× faster even at TV= 0.004.

Together, these findings suggest that ratio spread is a better predictor of deterministic algorithm performance than any property of TV distance itself, and that adaptive sampling is a practically effective modification to the randomized algorithm when distributions are expected to be well-separated.

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